

Aggregate Demand II: Applying the IS–LM Model

Chapter 14

Econ 313 – Intermediate Macroeconomics

University of Tennessee, Knoxville

July 30, 2026

IN THIS CHAPTER, YOU WILL LEARN: PART 1

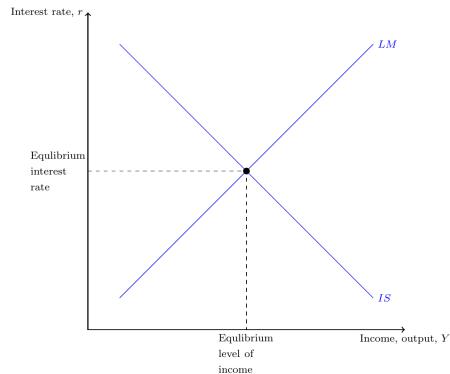
- ▶ Chapter 12 introduced the model of aggregate demand and supply.
- ▶ Chapter 13 developed the *IS-LM* model, the basis of the aggregate demand curve.

IN THIS CHAPTER, YOU WILL LEARN: PART 2

- ▶ How to use the $IS-LM$ model to analyze the effects of shocks, fiscal policy, and monetary policy
- ▶ How to derive the aggregate demand curve from the $IS-LM$ model
- ▶ Several theories about what caused the Great Depression

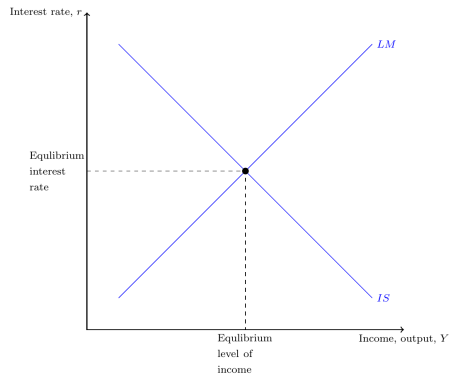
Equilibrium in the IS–LM model

- ▶ The IS curve represents equilibrium in the goods market.
- ▶ The LM curve represents money market equilibrium.
- ▶ The intersection determines the unique combination of Y and r that satisfies equilibrium in both markets.



Policy analysis with the IS–LM model

- ▶ We can use the *IS–LM* model to analyze the effects of
 - ▶ fiscal policy: G and/or T
 - ▶ monetary policy: M



Fiscal Policy: An increase in government purchases (1 of 5)

- ▶ An increase in government purchases shifts the IS curve to the right. The equilibrium moves from point A to point B . Income rises from Y_1 to Y_2 , and the interest rate rises from r_1 to r_2 .

Fiscal Policy: An increase in government purchases (2 of 5)

- ▶ The initial equilibrium is at A .
- ▶ An increase in government purchases causes the IS curve to shift out by

Fiscal Policy: An increase in government purchases (3 of 5)

- ▶ Resulting in a new equilibrium at point B .

Fiscal Policy: An increase in government purchases (4 of 5)

- ▶ Resulting in a higher level of income; Y_1 increases to Y_2 .

Fiscal Policy: An increase in government purchases (5 of 5)

- And a higher interest rate; r_1 increases to r_2 .

Fiscal Policy: A tax cut

- ▶ A decrease in taxes shifts the IS curve to the right. The equilibrium moves from point A to point B . Income rises from Y_1 to Y_2 , and the interest rate rises from r_1 to r_2 .

Monetary Policy: An increase in M (1 of 5)

- ▶ An increase in the money supply shifts the LM curve downward. The equilibrium moves from point A to point B . Income rises from Y_1 to Y_2 , and the interest rate falls from r_1 to r_2 .

Monetary Policy: An increase in M (2 of 5)

- ▶ An increase in the money supply shifts the LM curve downward.

Monetary Policy: An increase in M (3 of 5)

- ▶ The equilibrium moves from point A to point B .

Monetary Policy: An increase in M (4 of 5)

- ▶ Income rises from Y_1 to Y_2 .

Monetary Policy: An increase in M (5 of 5)

- And the interest rate falls from r_1 to r_2 .

Interaction between monetary and fiscal policy

Model:

- ▶ Monetary and fiscal policy variables (M , G , and T) are exogenous.
- ▶ Real world:
 - ▶ Monetary policymakers may adjust M in response to changes in fiscal policy or vice versa.
 - ▶ Such interactions may alter the impact of the original policy change.

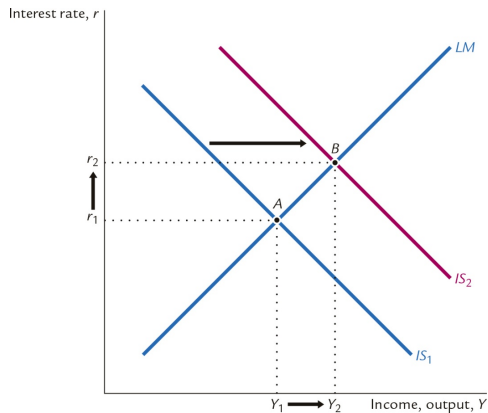
The Fed's response to $\Delta G > 0$

- ▶ Suppose Congress increases G .
- ▶ Possible Fed responses:
 - ▶ Hold M constant
 - ▶ Hold r constant
 - ▶ Hold Y constant
- ▶ In each case, the effects of ΔG are different . . .

▶ Numerical comparison

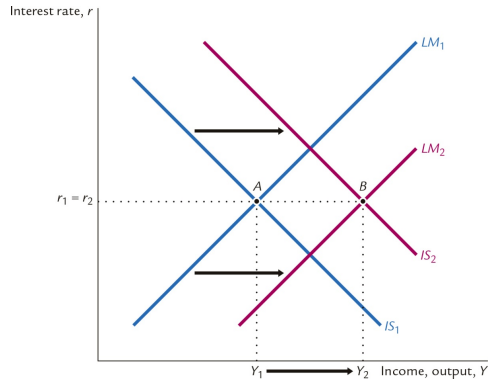
Response 1: Hold M constant

- ▶ An increase in G shifts the IS curve out. The Fed holds the money supply constant, and therefore, the LM curve stays the same. Both income and the interest rate rise.



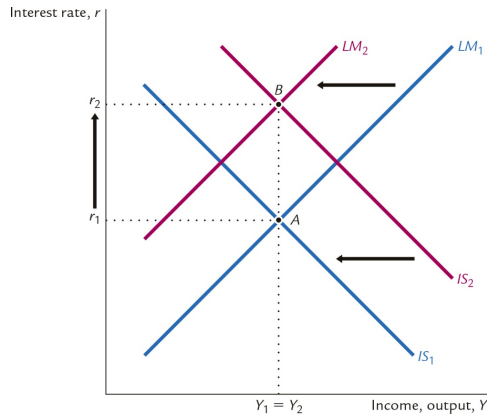
Response 2: Hold r constant

- An increase in G shifts the IS curve out. To keep r constant, the Fed then increases the money supply. Income increases by more than if the Fed held M constant.



Response 3: Hold Y constant

- An increase in G shifts the IS curve out. To keep Y constant, the Fed then decreases the money supply.



Shocks in the IS–LM model, part 1

- ▶ **IS shocks:** exogenous changes in the demand for goods and services
- ▶ For example:
 - ▶ stock market boom or crash → change in households' wealth → ΔC
 - ▶ change in business or consumer confidence or expectations → ΔI and/or ΔC

Shocks in the IS–LM model, part 2

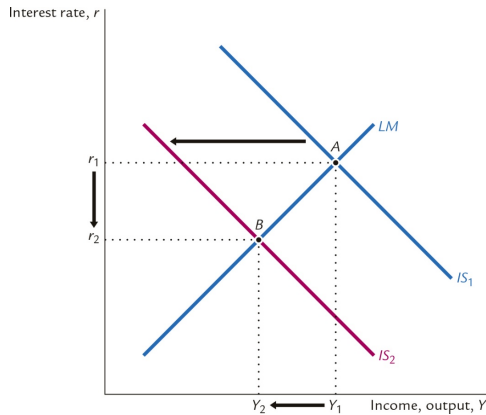
- ▶ **LM shocks:** exogenous changes in the demand for money
- ▶ For example:
 - ▶ During Covid-19, consumers and companies did not wish to handle cash and switched to more cashless methods of payments.
 - ▶ More ATMs or the internet reduce money demand.

NOW YOU TRY: Analyze shocks with the IS–LM model

1. Use the *IS–LM* model to analyze the effects of
 - ▶ a housing market crash that reduces consumers' wealth.
 - ▶ consumers using cash in transactions more frequently in response to an increase in identity theft.
2. For each shock,
 - ▶ use the *IS–LM* diagram to determine the effects on Y and r .
 - ▶ figure out what happens to C , I , and the unemployment rate.

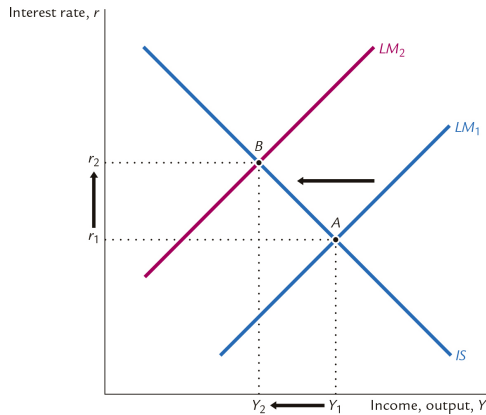
NOW YOU TRY: Analyze shocks with the IS–LM model, answer, part 1

1. IS shifts left, causing r and Y to fall.
2. C falls due to lower wealth and lower income.
3. I rises because r is lower.
4. u rises because Y is lower. (Okun's law)



NOW YOU TRY: Analyze shocks with the IS–LM model, answer, part 2

1. LM shifts left, causing r to rise and Y to fall.
2. C falls due to lower income.
3. I falls because r is higher.
4. u rises because Y is lower. (Okun's law)



CASE STUDY: The U.S. recession of 2001, part 1

During 2001:

- ▶ 2.1 million jobs lost, unemployment rose from 3.9 percent to 5.8 percent.
- ▶ GDP growth slowed to 0.8 percent (compared to 3.9 percent average annual growth during 1994–2000).



Alan Greenspan, Fed Chair
1987–2006. The Fed cut its target
rate from 6.5% to 1.75% during
2001.

CASE STUDY: The U.S. recession of 2001, part 2

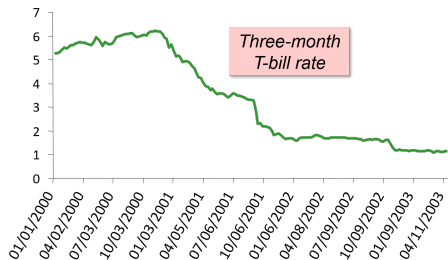
- ▶ Causes: 1) Stock market decline $\rightarrow \downarrow C$
 - ▶ S&P 500 fell from 1,500 in August 2000 to 1,150 in December 2001.
- ▶ Causes: 2) 9/11
 - ▶ increased uncertainty
 - ▶ fall in consumer and business confidence
 - ▶ result: lower spending, *IS* curve shifted left
- ▶ Causes: 3) Corporate accounting scandals
 - ▶ Enron, WorldCom, etc.
 - ▶ reduced stock prices, discouraged investment

CASE STUDY: The U.S. recession of 2001, part 3

- ▶ Fiscal policy response: shifted IS curve right
 - ▶ tax cuts in 2001 and 2003
 - ▶ spending increases
 - ▶ airline industry bailout
 - ▶ NYC reconstruction
 - ▶ Afghanistan war

CASE STUDY: The U.S. recession of 2001, part 4

- ▶ Monetary policy response: shifted LM curve right



What is the Fed's policy instrument? Part 1

- ▶ The news media commonly report the Fed's policy changes as interest rate changes, as if the Fed has direct control over market interest rates.
- ▶ In fact, the Fed **targets** the *federal funds rate*—the interest rate banks charge one another on overnight loans.
- ▶ The Fed changes the money supply and shifts the *LM* curve to achieve its target.
- ▶ Other short-term rates typically move with the federal funds rate.

What is the Fed's policy instrument? Part 2

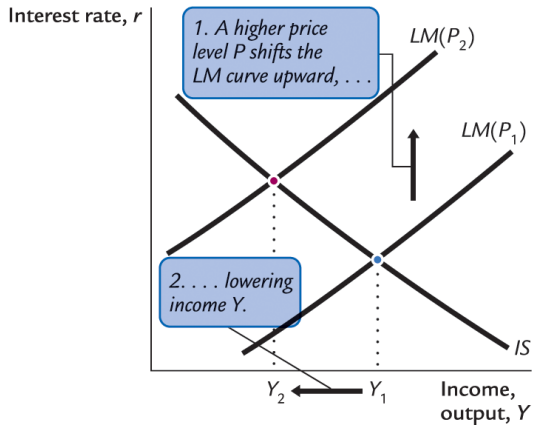
- ▶ Why does the Fed target interest rates instead of the money supply?
 - ▶ They are easier to measure than the money supply.
 - ▶ The Fed might believe that *LM* shocks are more prevalent than *IS* shocks. If so, targeting the interest rate stabilizes income better than targeting the money supply.

IS-LM and aggregate demand

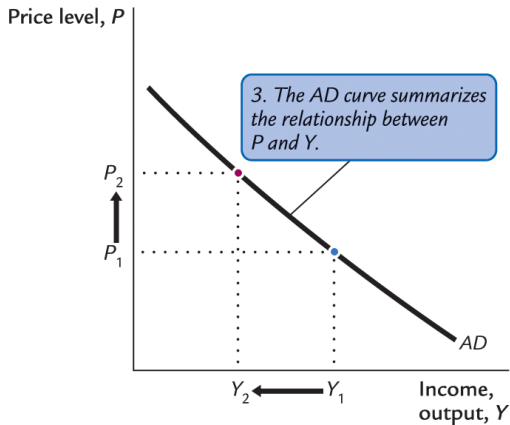
- ▶ So far, we've been using the *IS-LM* model to analyze the short run, when the price level is assumed to be fixed.
- ▶ However, a change in P would shift LM and would therefore affect Y .
- ▶ The **aggregate demand curve** (*introduced in Chapter 11*) captures this relationship between P and Y .

Deriving the AD curve

(a) The IS–LM Model



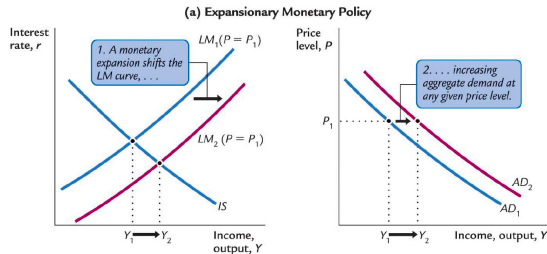
(b) The Aggregate Demand Curve



Monetary policy and the AD curve

The Fed can increase aggregate demand:

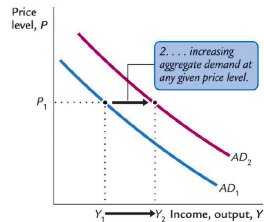
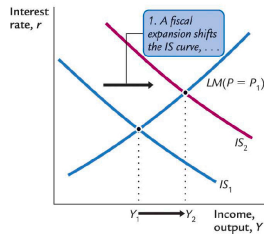
$\uparrow M \Rightarrow LM \text{ shifts right} \Rightarrow \downarrow r \Rightarrow \uparrow I \Rightarrow \uparrow Y$ at each value of P .



Fiscal policy and the AD curve

$\uparrow G$ and/or $\downarrow T \Rightarrow IS$ shifts right $\Rightarrow \uparrow Y$
at each value of P .

(b) Expansionary Fiscal Policy



IS–LM and AD–AS in the short run and in the long run (1 of 2)

Recall from Chapter 12: the force that moves the economy from the short run to the long run is the gradual adjustment of prices.

In the short-run equilibrium, if	then over time, the price level will
----------------------------------	--------------------------------------

$$Y > \bar{Y}$$

rise

$$Y < \bar{Y}$$

fall

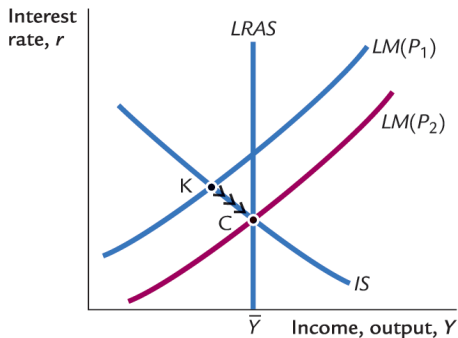
$$Y = \bar{Y}$$

remain constant

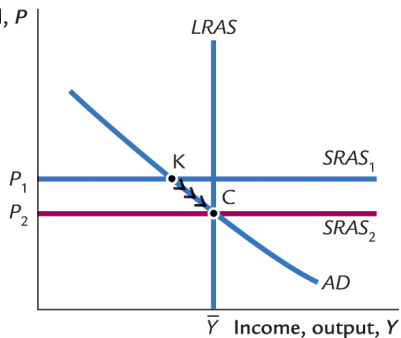
IS-LM and AD-AS in the short run and in the long run (2 of 2)

- Starting with $Y < \bar{Y}$, the price level will fall (from P_1 to P_2), causing both the LM and $SRAS$ curves to shift down. The short-run (Keynesian) equilibrium at K moves to the new long-run (Classical) equilibrium at C .

(a) The IS-LM Model

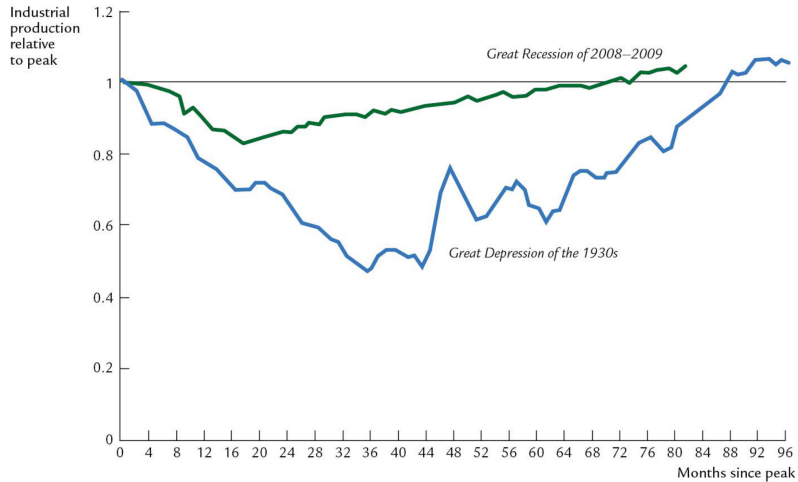


(b) The Model of Aggregate Supply and Aggregate Demand



Copyright © Worth Publishers

The Great Depression



THE SPENDING HYPOTHESIS: Shocks to the IS curve

- ▶ Asserts that the Depression was largely due to an exogenous fall in the demand for goods and services—a leftward shift of the *IS* curve.
- ▶ Evidence: Output and interest rates both fell, which is what a leftward *IS* shift would cause.

THE SPENDING HYPOTHESIS: Reasons for the IS shift

- ▶ Stock market crash reduced consumption.
 - ▶ October 1929–December 1929: S&P 500 fell 17 percent.
 - ▶ October 1929–December 1933: S&P 500 fell 71 percent.
- ▶ Drop in investment
 - ▶ Correction after overbuilding in the 1920s.
 - ▶ Widespread bank failures made it harder to obtain financing for investment.

THE MONEY HYPOTHESIS: A shock to the LM curve

- ▶ Asserts that the Depression was largely due to the huge fall in the money supply.
- ▶ Evidence: $M1$ fell 25 percent during 1929–1933.
- ▶ But, two problems with this hypothesis:
 - ▶ P fell even more, so M/P actually rose slightly during 1929–1931.
 - ▶ Nominal interest rates fell, which is the opposite of what a leftward LM shift would cause.

THE MONEY HYPOTHESIS AGAIN: The effects of falling prices, part 1

Asserts that the severity of the Depression was due to a huge deflation:

- ▶ P fell 25 percent during 1929–1933.
- ▶ This deflation was probably caused by the fall in M , so perhaps money played an important role after all.
- ▶ In what ways does a deflation affect the economy?

THE MONEY HYPOTHESIS AGAIN: The effects of falling prices, part 2

The stabilizing effects of deflation:

- ▶ $\downarrow P \rightarrow \uparrow(M/P) \rightarrow LM \text{ shifts right} \rightarrow \uparrow Y$
- ▶ **Pigou effect:**
 - ▶ $\downarrow P \rightarrow \uparrow(M/P)$
 - ▶ $\rightarrow \text{consumers' wealth } \uparrow$
 - ▶ $\rightarrow \uparrow C$
 - ▶ $\rightarrow IS \text{ shifts right}$
 - ▶ $\rightarrow \uparrow Y$

THE MONEY HYPOTHESIS AGAIN: The effects of falling prices, part 3

The destabilizing effects of expected deflation:

- ▶ $\downarrow E\pi$
 - ▶ $\rightarrow r \uparrow$ for each value of i
 - ▶ $\rightarrow I \downarrow$ because $I = I(r)$
 - ▶ $\rightarrow$ planned expenditure and aggregate demand $\downarrow$
 - ▶ $\rightarrow$ income and output $\downarrow$

THE MONEY HYPOTHESIS AGAIN: The effects of falling prices, part 4

- ▶ The destabilizing effects of unexpected deflation: **debt-deflation theory**
- ▶ $\downarrow P$ (if unexpected)
 - ▶ $\rightarrow$ transfers purchasing power from borrowers to lenders
 - ▶ $\rightarrow$ borrowers spend less, lenders spend more
 - ▶ $\rightarrow$ if borrowers' propensity to spend is larger than lenders', then aggregate spending falls, the IS curve shifts left, and Y falls

Why another Depression is unlikely

Policymakers (or their advisers) now know much more about macroeconomics:

- ▶ The Fed knows better than to let M fall so much, especially during a contraction.
 - ▶ Fiscal policymakers know better than to raise taxes or cut spending during a contraction.
- ▶ Federal deposit insurance makes widespread bank failures very unlikely.
- ▶ Automatic stabilizers make fiscal policy expansionary during an economic downturn.

CASE STUDY: The 2008–2009 financial crisis and recession (1 of 2)

- ▶ 2009: Real GDP fell; unemployment rate approached 10 percent.
- ▶ Important factors in the crisis:
 - ▶ early 2000s Federal Reserve interest rate policy
 - ▶ subprime mortgage crisis
 - ▶ bursting of house price bubble, rising foreclosure rates
 - ▶ falling stock prices
 - ▶ failing financial institutions
 - ▶ declining consumer confidence, drop in spending on consumer durables and investment goods

CASE STUDY: The 2008–2009 financial crisis and recession (2 of 2)

Fiscal Policy:

- ▶ American Recovery and Reinvestment Act (ARRA) was \$787 billion passed in early 2009.
 - ▶ Yet, later in 2010–2014, government spending decreased due to austerity measures (over a \$300 billion drop).
- ▶ Monetary Policy:
 - ▶ Lowered the federal funds rate to zero
 - ▶ Engaged in nontraditional monetary policy (forward guidance and quantitative easing)

Liquidity trap and unconventional monetary policy

- ▶ **Liquidity trap:** A situation where interest rates have fallen to zero, and therefore, it is possible that (conventional) monetary policy is no longer effective. At times referred to as the “zero lower bound.”
- ▶ **Unconventional monetary policies:**
 - ▶ **Forward guidance:** a policy of announcing *future* monetary actions, for example, “we expected . . . to maintain [low interest rates] until labor market conditions have reached levels consistent with maximum employment . . .”
 - ▶ **Quantitative easing (QE):** buying of long-term government debt, mortgage-backed securities, corporate debt, state and local debt, etc.

CHAPTER SUMMARY, PART 1

- ▶ *IS-LM* model
 - ▶ a theory of aggregate demand
 - ▶ exogenous: M, G, T, P exogenous in short run, Y in long run
 - ▶ endogenous: r, Y endogenous in short run, P in long run
 - ▶ *IS* curve: goods market equilibrium
 - ▶ *LM* curve: money market equilibrium

CHAPTER SUMMARY, PART 2

- ▶ *AD* curve
 - ▶ shows relationship between P and the *IS*–*LM* model's equilibrium Y .
 - ▶ negative slope because $\uparrow P \rightarrow \downarrow (M/P) \rightarrow \uparrow r \rightarrow \downarrow I \rightarrow \downarrow Y$
 - ▶ expansionary fiscal policy shifts *IS* curve right, raises income, and shifts *AD* curve right.
 - ▶ expansionary monetary policy shifts *LM* curve right, raises income, and shifts *AD* curve right.
 - ▶ *IS* or *LM* shocks shift the *AD* curve.

Appendix: The Fed's three responses — numerical comparison

Baseline model. From Chapter 13:

- ▶ $Y^{IS} = 1350 - 50 r$
- ▶ $Y^{LM} = 1000 + 125 r$ (given $M/P = 400$)
- ▶ Equilibrium: $r^* = 2\%$, $Y^* = 1250$.

Shock. Congress raises G by 100; IS shifts to

$$Y^{IS'} = 1600 - 50 r.$$

Now trace out three possible Fed reactions.

◀ back

Appendix — Response 1: hold M constant

M/P stays at 400, so LM does not move.

$$Y^{IS'} = Y^{LM}: \quad 1600 - 50r = 1000 + 125r \implies r^* = \frac{600}{175} \approx 3.43\%.$$

$$Y^* = 1600 - 50(3.43) \approx 1428.5.$$

Compared to baseline: $\Delta r \approx +1.43\text{pp}$, $\Delta Y \approx +178.5$.

Partial crowding out: $\Delta Y \approx 178.5$, less than the Keynesian-cross prediction of $\Delta Y = 250$ because higher r reduces I .

◀ back

Appendix — Response 2: hold r constant

Fed accommodates: keeps r at 2% by raising M (shifting LM right).

Interest rate is fixed: substitute $r = 2$ into new IS:

$$Y^* = 1600 - 50(2) = 1500.$$

$\Delta Y = +250$ (full Keynesian multiplier, no crowding out).

Required M/P : at $r = 2$, $Y = 1500$, money demand is $0.4(1500) - 50(2) = 500$. So M/P must rise from 400 to 500.

Fiscal stimulus + accommodative monetary policy delivers the maximum Y boost. This is what Bernanke did in 2008–09 alongside the ARRA stimulus.

◀ back

Appendix — Response 3: hold Y constant

Fed leans against fiscal expansion: keeps Y at 1250 by contracting M .

From new IS: $1250 = 1600 - 50r \implies r^* = 7\%$.

Required M/P : $0.4(1250) - 50(7) = 500 - 350 = 150$. So M/P must fall from 400 to 150.

	Y	r	M/P
Baseline	1250	2%	400
R1 (hold M)	1428.5	3.4%	400
R2 (hold r)	1500	2%	500
R3 (hold Y)	1250	7%	150

Same fiscal shock, three very different outcomes for r and investment — depending on the Fed's reaction. This is why understanding monetary–fiscal coordination matters.

◀ back